FROM FIELDS TO CIRCUITS

Recovering $R(x)$ and $C(x)$ from a TDR Trace

Capstone Project Report — Problem B: Transmission Line Signal Integrity

Team (Group 4): Arya P R, Jophin John, Meeval Varghese, Arjun S Bijoy, Bevin Rari

Department of Electronics and Communication Engineering
Saintgits College of Engineering
2026

Abstract

Time-domain reflectometry (TDR) is widely used to find hidden faults in coaxial cables and printed-circuit traces without cutting them open, but the reflected voltage measured at the input is only an indirect record of how the line's distributed parameters vary along its length. This report studies the inverse problem of recovering the resistance profile $R(x)$ and the capacitance profile $C(x)$ of a 1 m, 50 Ω line from a single noisy reflection trace. The line is described by the telegrapher's equations, simulated with a staggered finite-difference time-domain (FDTD) scheme, and linearised into a sensitivity matrix that links parameter changes to changes in the measured voltage. An ℓ_1 linear program, relinearised three times, then estimates the profiles. In a synthetic test with a hidden fault near $x = 0.5$ m ($R \times 4$ and $C \times 0.6$ over three nodes) and 2 mV of noise, the method recovered the capacitance dip to 59 pF/m against a true value of 60 pF/m, and the fault location stayed within about half a grid cell over eight noise seeds. At 5 mV noise the worst error rose to 1.7 cells, which still meets the two-cell target. The resistance spike could not be recovered, because TDR is roughly 2600 times less sensitive to series resistance than to capacitance in this setup. The study therefore shows both where the method works and where a single TDR trace simply does not carry enough information.

1. Introduction

A transmission line is rarely as uniform as the diagrams in a textbook suggest. A coaxial cable that has been bent too sharply, crushed under a clamp or soaked with moisture, or a PCB trace that passes a damaged via, will show position-dependent changes in its series resistance R, series inductance L, shunt conductance G and shunt capacitance C. These changes are hidden inside the line, and in a practical TDR measurement the only quantity we can observe is the voltage reflected back to the input after a fast step has been sent down the line.

The problem matters because every change in local impedance reflects part of the signal. In high-speed links those reflections distort edges, shift timing and eventually cause bit errors. TDR is attractive because it can locate such a fault from one end of the cable without cutting it, but turning the measured echo into an actual map of the line is not straightforward. Different R and C profiles can produce very similar echoes, and measurement noise makes a direct inversion unstable.

Our approach was to build the physics first and the inversion second. We model the line with the telegrapher's equations, solve them numerically with an FDTD scheme to obtain a forward model of the TDR trace, and perturb each parameter in turn to obtain a linear system $A\,\mathbf{p} = \mathbf{b}$ that relates parameter changes to changes in the trace. The linear system is solved with an ℓ_1 linear program, and because a 40% change in capacitance is not small, the linearisation is repeated three times around the latest estimate. The project target was a line of length $\ell = 1$ m with 50

CamScanner

spatial nodes, nominal values $R_0 = 0.1\,\Omega/\text{m}$, $L_0 = 250$ nH/m, $G_0 = 0$ and $C_0 = 100$ pF/m (so that $Z_0 = \sqrt{L_0/C_0} \approx 50\,\Omega$), and a hidden fault near $x = 0.5$ m in which R is multiplied by 4 and C by 0.6 over three nodes. The goal was to recover $R(x)$, $C(x)$ and the fault location to within two grid cells despite small measurement noise.

2. Related Work

The inverse problem of finding an impedance profile from reflection data is not new. Bruckstein and Kailath [3] studied inverse scattering for discrete transmission-line models and showed how the impedance of a lossless line can be peeled off layer by layer from its reflection response. In the absence of noise the method is exact, but each layer is computed from the one before it, so errors accumulate along the line and a real measurement with noise eventually spoils the deeper layers. Layer peeling also assumes a lossless line, whereas real cables always have some series resistance.

A second line of work concerns how to fit data that contain outliers. Claerbout and Muir [4] introduced robust fitting of erratic data by minimising the sum of absolute residuals instead of squared residuals, so a few bad samples cannot dominate the fit. More recently, Candès, Romberg and Tao [5] showed that ℓ_1 minimisation can recover sparse signals from a small number of measurements, which is exactly the situation of a single localised fault in a long line. The forward model rests on standard texts: Pozar [1] gives the line theory behind the telegrapher's equations, and Taflove and Hagness [2] describe the finite-difference time-domain method that we use to simulate the line.

Several gaps remain. A TDR trace is an indirect measurement, so different R and C profiles can give similar echoes, and noise makes direct inversion unstable. Loss is often ignored in inversion studies, and the effect of noise on the accuracy of the fault position is rarely quantified in units that a practitioner can use, such as grid cells. This project combines a lossy physics model with an ℓ_1 linear program and reports honestly both where the recovery works and where it fails.

3. Methodology

This section follows the chain of reasoning from the physics of the line to the optimisation problem that is finally solved. Each equation is introduced because the next step needs it, and every symbol is explained where it first appears.

3.1 The distributed line model

A thin section of line is treated as a small loop. Faraday's law, $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$, integrated over this loop with Stokes' theorem relates the change of voltage along the line to the magnetic flux linking the loop and to the voltage dropped across the conductor resistance. Ampère's law, $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D}/\partial t$, applied to the gap between the conductors relates the change of current along the line to the conduction current that leaks across the dielectric and the displacement current that charges the capacitance. Together they give the telegrapher's equations:

$$\frac{\partial V}{\partial x} = -R(x)\, I - L(x)\, \frac{\partial I}{\partial t}, \tag{1}$$

$$\frac{\partial I}{\partial x} = -G(x)\, V - C(x)\, \frac{\partial V}{\partial t}. \tag{2}$$

Here $V(x,t)$ is the voltage between the conductors and $I(x,t)$ is the current along them, both functions of position x and time t. The four per-unit-length parameters carry the physics: $R(x)$

CamScanner

in Ω/m is the series resistance that dissipates energy, $L(x)$ in H/m stores magnetic energy, $G(x)$ in S/m represents leakage through the dielectric, and $C(x)$ in F/m stores electric energy. Writing them as functions of x is the key step for this project, because a fault is simply a place where one of them departs from its nominal value.

3.2 Wave behaviour, impedance and the origin of the echo

To understand how a fault produces an echo, we transform the equations in time with the Laplace variable s. For a uniform section the two first-order equations combine into a single second-order equation, and the section is then described by two quantities, the propagation constant and the characteristic impedance:

$$\gamma(s) = \sqrt{(R + sL)(G + sC)}, \tag{3}$$

$$Z_0(s) = \sqrt{\frac{R + sL}{G + sC}}. \tag{4}$$

In these expressions s is the complex frequency, so $R + sL$ is the series impedance per metre and $G + sC$ is the shunt admittance per metre. The propagation constant $\gamma(s)$ tells us how a wave changes as it travels, with its real part giving attenuation and its imaginary part giving phase, while $Z_0(s)$ is the ratio of voltage to current for a wave travelling in one direction. The transformed voltage $\hat{V}(x, s)$ obeys

$$\frac{d^2\hat{V}}{dx^2} = \gamma^2\hat{V}, \qquad \hat{V}(x, s) = V^+e^{-\gamma x} + V^-e^{+\gamma x}, \tag{5}$$

which has two travelling solutions, a forward wave $V^+e^{-\gamma x}$ and a backward wave $V^-e^{+\gamma x}$. V^+ and V^- are the amplitudes of the forward and backward waves. A fault generates the backward wave: the forward wave arrives at a point where the line properties change and part of it turns back toward the source.

For our nominal line the resistance is tiny compared with the reactance, so the expressions simplify to

$$Z_0 \approx \sqrt{\frac{L_0}{C_0}}, \qquad v = \frac{1}{\sqrt{LC}}. \tag{6}$$

With $L_0 = 250$ nH/m and $C_0 = 100$ pF/m this gives $Z_0 \approx 50\,\Omega$ and a wave speed $v = 1/\sqrt{LC} \approx 2 \times 10^8$ m/s, about two thirds of the speed of light. When the loss terms are kept, the voltage satisfies the damped wave equation

$$\frac{\partial^2 V}{\partial x^2} = LC\,\frac{\partial^2 V}{\partial t^2} + (RC + GL)\,\frac{\partial V}{\partial t} + RG\,V. \tag{7}$$

This is a hyperbolic equation of the wave type, with the first term on the right-hand side setting the propagation speed through LC, the middle term $(RC + GL)$ acting as a damping coefficient and the last term RG a small attenuation that vanishes when either R or G is zero. The equation is written for constant coefficients; in the real line the coefficients vary with x, and the first-order system (1)–(2) is what we actually solve.

The echo itself is governed by the reflection coefficient at an impedance step, where Z_1 is the impedance seen by the wave before the step and Z_2 the impedance after it:

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}. \tag{8}$$

A fault in which C drops to 0.6 of its nominal value raises the local impedance to about $50/\sqrt{0.6} \approx 64.5\,\Omega$, so $\Gamma \approx 0.13$ at the entry to the fault. With a launched step of 0.5 V this

CamScanner

predicts an echo of roughly 60–65 mV, which is consistent in size with the echo we see in the results. The position of the fault follows from the arrival time of the echo, because the pulse has to travel to the fault and back:

$$t_{\text{echo}} = \frac{2x}{v}, \tag{9}$$

where t_{echo} is the round-trip delay and x is the distance to the fault. An echo at 5 ns therefore places the impedance change at $x = v\,t_{\text{echo}}/2 = 0.5$ m. The same reasoning also explains why resistance is hard to see. A series resistance changes the impedance by an amount of order ΔR against $50\,\Omega$, so a tiny added resistance produces a much smaller reflection than the large relative change in C does.

3.3 Initial and boundary conditions

The first-order system needs conditions at the ends of the line and at the start of time. At the input, a 1 V step source $V_s(t)$ drives the line through a source resistance $R_s = 50\,\Omega$, and at the far end the line is terminated in a load $R_L = 50\,\Omega$:

$$V(0,t) = V_s(t) - R_s\,I(0,t), \qquad V(\ell,t) = R_L\,I(\ell,t). \tag{10}$$

The first relation says that the voltage at the input is the source voltage minus the drop across R_s, and the second says that the far end behaves as a resistor. Since R_L equals the nominal Z_0, the load reflection coefficient is zero and the far end does not reflect, so any echo in the trace belongs to the fault. The line starts unenergised, which we write as

$$V(x,0) = 0, \qquad I(x,0) = 0, \tag{11}$$

meaning that no voltage or current exists anywhere on the line before the step arrives. Because the source resistance and the line impedance are both $50\,\Omega$, the step launched onto the line has an amplitude of 0.5 V.

3.4 Finite-difference time-domain discretisation

The equations are solved on a staggered grid of $N_x = 50$ nodes over $\ell = 1$ m, where voltage lives on the nodes and current lives halfway between them. This arrangement lets each spatial derivative be a centred difference over a single cell, which is accurate and cheap. The grid spacing and the stability condition are

$$S = \frac{v\,\Delta t}{\Delta x} \le 1, \qquad \Delta x = \frac{\ell}{N_x - 1}, \tag{12}$$

where Δx is the cell size (about 0.0204 m), Δt is the time step (0.05 ns) and S is the Courant number, which must not exceed one for the explicit scheme to remain stable. For our line $S \approx 0.5$, comfortably inside the stable range. The updates for voltage and current alternate in time:

$$V_i^{n+1} = V_i^n - \frac{\Delta t}{C_i\,\Delta x}\left(I_{i+1/2}^{n+1/2} - I_{i-1/2}^{n+1/2}\right) - \frac{G_i\,\Delta t}{C_i}\,V_i^n, \tag{13}$$

$$I_{i+1/2}^{n+3/2} = I_{i+1/2}^{n+1/2} - \frac{\Delta t}{L_{i+1/2}\,\Delta x}\left(V_{i+1}^{n+1} - V_i^{n+1}\right) - \frac{R_{i+1/2}\,\Delta t}{L_{i+1/2}}\,I_{i+1/2}^{n+1/2}. \tag{14}$$

The superscript n marks the time step and the subscript i the node, with half-integer indices marking the staggered positions where current is stored. The first update says that the voltage at node i changes because the currents entering and leaving the node differ, with C_i turning that charge imbalance into a voltage change, and with G_i draining a little charge. The second says

CamScanner

that the current in a cell changes because of the voltage difference across it, with L setting how quickly, while the last term, containing R, dissipates energy. The values of R and L needed at the half-node positions are taken from the neighbouring nodes. Running the loop with the source and load conditions of equation (10) produces the voltage at the input $V_{\text{in}}(t)$, which is the TDR trace.

3.5 The inverse problem

The forward model maps a parameter vector to a TDR trace. Let $\mathbf{p}$ hold the 100 unknowns, the 50 values of R followed by the 50 values of C, each expressed relative to its nominal value so that $\mathbf{p} = 1$ means nominal. Let $\mathbf{y}(\mathbf{p})$ be the 240 simulated samples of the trace (12 ns at 0.05 ns). Near a current estimate $\mathbf{p}_0$ the trace can be approximated linearly:

$$\mathbf{y}(\mathbf{p}) \approx \mathbf{y}(\mathbf{p}_0) + A\,(\mathbf{p} - \mathbf{p}_0) \;\Rightarrow\; A\,(\mathbf{p} - \mathbf{p}_0) \approx \mathbf{b}, \quad \mathbf{b} = \mathbf{y}_{\text{meas}} - \mathbf{y}(\mathbf{p}_0). \tag{15}$$

Here A is the sensitivity matrix, $\mathbf{y}_{\text{meas}}$ is the measured trace and $\mathbf{b}$ is the difference between the measurement and the trace predicted by the current estimate. The matrix A is built column by column by perturbing one parameter at a time and re-simulating:

$$A_{kj} = \frac{y_k(\mathbf{p}_0 + \delta\, p_{0,j}\, \mathbf{e}_j) - y_k(\mathbf{p}_0)}{\delta\, p_{0,j}}, \qquad \delta = 0.02, \tag{16}$$

where A_{kj} is the change of the k-th time sample caused by a change of the j-th parameter, $\mathbf{e}_j$ is the unit vector that selects parameter j, $p_{0,j}$ is its current value, and $\delta = 0.02$ is the 2% perturbation size. The resulting matrix has 240 rows and 100 columns. Because the columns of A for neighbouring nodes are very similar and the data are noisy, inverting A directly would amplify the noise badly, so we use a regularised ℓ_1 formulation that can be solved as a linear program [4, 5, 6]:

$$\begin{aligned} \min_{\mathbf{p},\,\varepsilon,\,u} \; & \sum_{k=1}^{240} \varepsilon_k + \lambda \sum_{j=1}^{100} w_j\, u_j \\ \text{s.t.} \; & -\varepsilon \le A(\mathbf{p} - \mathbf{p}_0) - \mathbf{b} \le \varepsilon, \\ & -u \le \mathbf{p} - \mathbf{p}_0 \le u, \quad \mathbf{p} \ge 0. \end{aligned} \tag{17}$$

In this formulation ε_k is an auxiliary variable that bounds the absolute value of the k-th residual, so the objective $\sum \varepsilon_k$ equals the sum of absolute residuals, which resists outlying noise samples. The auxiliary variable u_j bounds $|p_j - p_{0,j}|$, so the second term is the ℓ_1 size of the change; the weight λ sets how strongly the solver prefers a sparse, localised change, which is what a single fault looks like. The weights w_j are larger for the R columns, because R is almost invisible to TDR and would otherwise absorb noise. The constraint $\mathbf{p} \ge 0$ keeps the parameters physical; since C falls at the fault and $\mathbf{p}$ is the relative parameter value itself, this causes no sign difficulty. The problem has several hundred variables and is solved with the HiGHS solver in SciPy's `linprog`. After each solution the estimate $\mathbf{p}_0$ is replaced by the new $\mathbf{p}$, the model is re-simulated, A and $\mathbf{b}$ are rebuilt, and the program is solved again. Three such passes are used, because the linear approximation is poor for a 40% change in capacitance but improves as the estimate approaches the truth.

3.6 Performance metrics

Three quantities are used to judge the result, all collected in one expression:

$$E_{\text{loc}} = \frac{|\hat{x}_f - x_f|}{\Delta x}, \qquad \text{RMS}_{\text{res}} = \sqrt{\frac{1}{K}\sum_{k=1}^{K} r_k^2}, \qquad \text{RMS}_C = \frac{1}{C_0}\sqrt{\frac{1}{N_x}\sum_{i=1}^{N_x} (\hat{C}_i - C_i)^2}. \tag{18}$$

CamScanner

The first, E_{loc}, is the location error in grid cells, where $\hat{x}_f$ is the fault position read from the recovered C profile and x_f is the true position. The second, RMS_{res}, is the root-mean-square of the residual samples r_k (measured trace minus model trace) over the $K = 240$ time samples; a good fit should leave a residual close to the noise level. The third, RMS_C, is the root-mean-square error of the recovered capacitance profile, where $\hat{C}_i$ and C_i are the estimated and true capacitance at node i, normalised by C_0. We also record the minimum recovered capacitance, which measures how deep the fault dip is.

4. Experimental Setup

All computations were done in Python 3. NumPy carries the arrays and the FDTD loop, SciPy supplies the linear programming solver (`linprog` with the HiGHS method), and Matplotlib produces the plots. No special hardware or commercial simulation software was needed. The physical and numerical settings are collected in Table 1.

Table 1: *Physical and numerical settings of the simulation study.*

Quantity	Value
Line length and nodes	$\ell = 1$ m, $N_x = 50$ ($\Delta x \approx 0.0204$ m)
Nominal R_0, L_0, G_0, C_0	$0.1\,\Omega$/m, 250 nH/m, 0 S/m, 100 pF/m
Terminations	$R_s = 50\,\Omega$ source, $R_L = 50\,\Omega$ load (matched)
Excitation	1 V step, 0.2 ns rise time
Time step and window	$\Delta t = 0.05$ ns, 12 ns (240 samples), Courant ≈ 0.5
Hidden fault	nodes 24–26 ($x \approx 0.49$–0.53 m), $R \times 4$, $C \times 0.6$
Noise	Gaussian, $\sigma = 2$ mV (1–5 mV tested)
Sensitivity matrix	240×100, 2% perturbation of each node
Inversion	ℓ_1 linear program, 3 relinearisation passes

The workflow is reproducible from these settings. First the baseline (nominal) and faulted TDR traces are simulated with the FDTD model and Gaussian noise is added to the faulted trace. Then the sensitivity matrix is built by perturbing each of the 100 parameters by 2% and re-simulating, which costs about 200 forward simulations per Jacobian. The linear program is solved three times, with the model and matrix rebuilt after each pass, and the $R(x)$ and $C(x)$ profiles and the residuals are then computed. Finally, to test how stable the fault location is, the whole procedure is repeated for eight different noise seeds at each noise level of 0, 1, 2 and 5 mV.

5. Results

5.1 The synthetic TDR response

Figure 1 shows the simulated input voltage for the nominal line and for the faulted line with 2 mV noise. After the step arrives, both traces settle near 0.5 V, which is the expected value when a 1 V source with $50\,\Omega$ resistance drives a $50\,\Omega$ line. A short burst of ringing appears right at the start of the step, and the nominal trace shows small ripples towards the end of the window. These features are present in both traces, so they cancel when the baseline is subtracted and do not influence the fault estimate.

The key feature is the positive echo near 5 ns, which rises to roughly 0.57 V, about 70 mV above the 0.5 V plateau and well above the 2 mV noise. Applying equation (9), a round trip of 5 ns at $v \approx 2 \times 10^8$ m/s puts the impedance change at about 0.5 m, matching the true fault position. The echo is positive because the reduced capacitance raises the local impedance above $50\,\Omega$, so

CamScanner

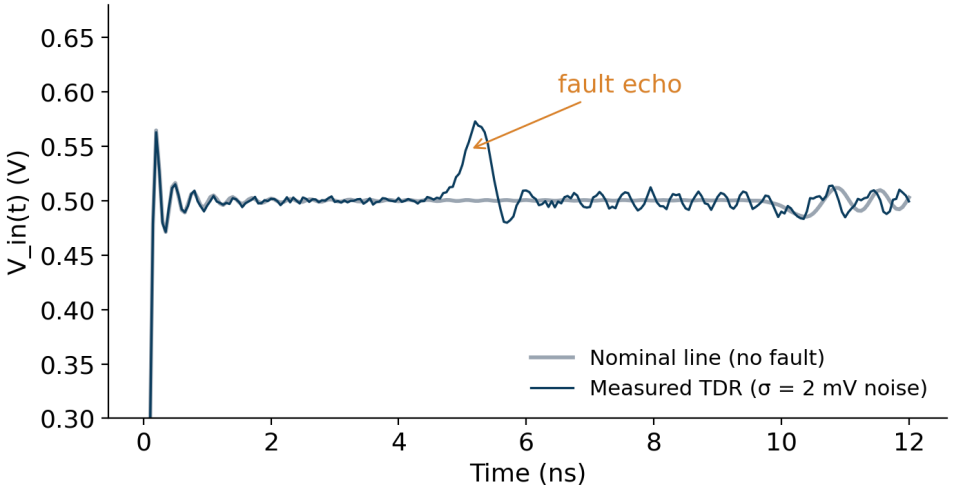


Figure 1: *Simulated TDR trace: nominal line (grey) and faulted line with 2 mV noise (dark). The fault echo appears near 5 ns. Simulated with the FDTD model; not a measured experiment.*

the reflection coefficient of equation (8) is positive. The echo is followed by a short negative swing as the wave leaves the faulty section and the impedance returns to normal.

5.2 Recovered $R(x)$ and $C(x)$ profiles

Figure 2 compares the true and recovered profiles after the three ℓ_1 passes. The capacitance panel shows a sharp, narrow dip at the position of the fault, with the recovered minimum at about 59 pF/m against the true 60 pF/m, and the recovered profile follows the nominal 100 pF/m closely away from the fault. Small wiggles of a pF/m or two remain along the line, and they are somewhat larger near the far end, which is what we would expect from noise that the sparsity term has not fully suppressed.

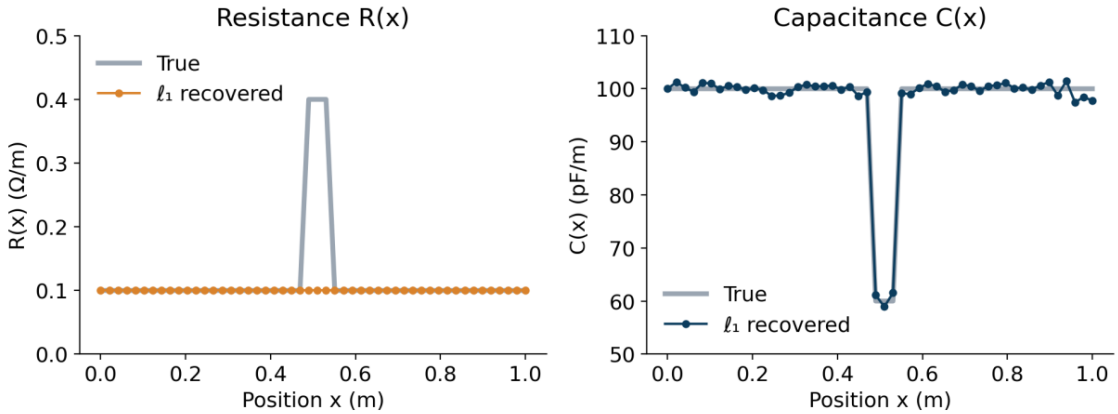


Figure 2: *True and ℓ_1-recovered resistance $R(x)$ (left) and capacitance $C(x)$ (right) at 2 mV noise.*

The resistance panel tells a different story. The true profile contains a spike to $0.4\,\Omega$/m at the fault, but the recovered profile stays flat at the nominal $0.1\,\Omega$/m. This is not a bug in the optimiser but a property of the measurement. Over three nodes the spike adds only about $0.02\,\Omega$ of series resistance, to be compared with a characteristic impedance of $50\,\Omega$, so its reflection is minute. The computed sensitivity of the trace to R is about 2593 times smaller than its sensitivity to C, which places the effect of the resistance spike far below the 2 mV noise. Together

CamScanner

with the large penalty that the solver places on the R columns, this means that the optimiser correctly refuses to attribute any of the echo to resistance. The fault is still located through the capacitance, which also changes the impedance.

5.3 Residuals and validation metrics

Figure 3 plots the residual, the measured trace minus the model trace, before and after inversion. Before inversion the nominal model cannot explain the data: the residual has a large peak of about 70 mV near 5 ns, a negative swing of about 20 mV after it, and further oscillations of the order of 10–20 mV later in the window. These later oscillations are far larger than the 2 mV noise, which points to secondary reflections caused by the fault that the nominal model does not contain. After inversion the residual is flat and fluctuates only at the few-millivolt level, in line with the noise, which means that the recovered profile reproduces the measured trace to within the noise. The residual RMS after inversion is 1.8 mV for a noise level of 2 mV, and the capacitance profile has an RMS error of 0.9% of C_0.

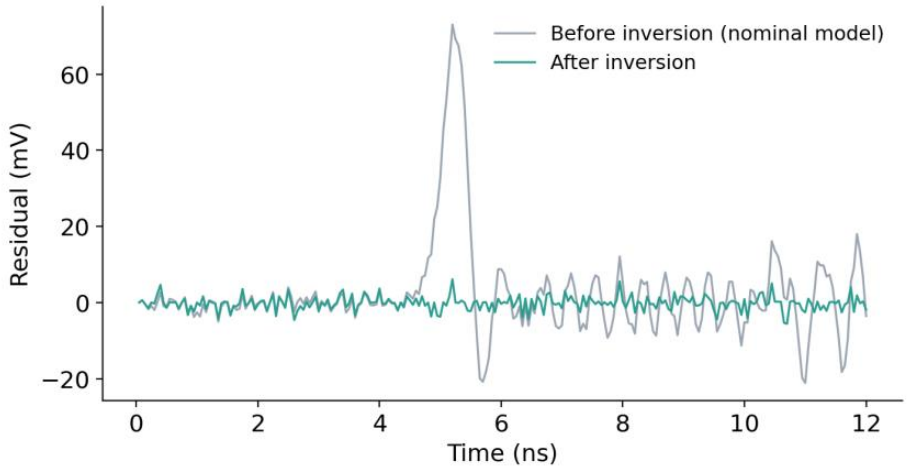


Figure 3: *Residual (measured minus model) before inversion with the nominal model and after inversion, at 2 mV noise.*

Table 2 summarises the stability of the fault location over eight noise seeds at each noise level, with the location error expressed in grid cells.

Table 2: *Fault-location error ($|estimated - true|$ in grid cells) and minimum recovered capacitance, 8 noise seeds per level.*

Noise σ	Mean error (cells)	Worst error (cells)	Min C ($\times C_0$)
0 mV	0.27	0.27	0.59
1 mV	0.30	0.50	0.59
2 mV	0.27	0.47	0.59
5 mV	0.68	1.69	0.57

With no noise at all, every seed gives the same answer, and the error is 0.27 cells for both the mean and the worst case. This offset therefore comes from the discretisation and the linearisation, not from noise, and it sets the floor for the method. At 1 mV and 2 mV noise the mean error stays near 0.3 cells and the worst case is about half a cell. The slightly smaller values at 2 mV than at 1 mV should not be read as an improvement from more noise; with only eight seeds per level, such small differences are within the statistical scatter. At 5 mV the picture changes,

CamScanner

with the mean error rising to 0.68 cells and the worst case to 1.69 cells, showing that the echo is beginning to sink into the noise. The minimum recovered capacitance stays at 0.59 of nominal for noise up to 2 mV and drops slightly to 0.57 at 5 mV, always close to the true factor of 0.6.

Against the success criterion, the fault was located within two grid cells at every noise level, and within one grid cell up to 2 mV; at 5 mV the worst case of 1.69 cells exceeds one cell but still meets the two-cell target.

6. Limitations

The most important limitation is that $R(x)$ is practically unidentifiable from a single TDR trace, so the resistance spike was not recovered. The study also relies on a linearised model, and the linearisation is inaccurate for a 40% change in capacitance; the three relinearisation passes are needed to compensate, and the method would need more care for larger faults. Finite bandwidth, timing uncertainty and noise limit the accuracy, and at 5 mV noise the worst-case error grows to about 1.7 cells.

All results come from synthetic data, produced by the same type of model used for the inversion, so the study does not test model mismatch. The inductance and conductance are assumed known, the terminations are assumed matched, and dispersion and connector effects are ignored. Finally, the regularisation weight λ and the trust-region limits that bound each update were tuned by hand, and each Jacobian requires about 200 forward simulations, which makes the method slower than a one-shot inversion. We reduced these risks by using a known synthetic ground truth, testing several noise seeds, and checking the residuals, but they remain open.

7. Conclusion and Future Scope

This project started from Maxwell's curl equations, derived the telegrapher's equations and used the Laplace transform to obtain the propagation constant $\gamma(s)$ and the characteristic impedance $Z_0(s)$. With an FDTD forward model and a sensitivity matrix, an ℓ_1 linear program with three relinearisation passes located a hidden fault to within about half a grid cell in the worst of eight seeds at 2 mV noise, and to under two cells at 5 mV, while recovering a sharp dip in $C(x)$ with a minimum close to the true 60 pF/m. The resistance profile could not be recovered, because TDR is far less sensitive to series loss than to capacitance. The work links electromagnetic modelling to a practical signal-integrity diagnostic and shows clearly which quantities a single trace can and cannot reveal.

Future work could resolve R and G by combining several TDR or S-parameter measurements, fit the propagation constant $\gamma(s)$ in the frequency domain, test the method on real cable and PCB data, use total-variation priors to handle several faults, and choose λ automatically instead of by hand.

References

[1] D. M. Pozar, *Microwave Engineering*, 4th ed. Wiley, 2011.

[2] A. Taflove and S. C. Hagness, *Computational Electrodynamics: The FDTD Method*, 3rd ed. Artech House, 2005.

[3] A. M. Bruckstein and T. Kailath, "Inverse scattering for discrete transmission-line models," *SIAM Review*, vol. 29, no. 3, pp. 359–389, 1987.

CamScanner

[4] J. F. Claerbout and F. Muir, "Robust modeling of erratic data," *Geophysics*, vol. 38, no. 5, pp. 826–844, 1973.

[5] E. J. Candès, J. Romberg and T. Tao, "Robust uncertainty principles," *IEEE Trans. Inf. Theory*, vol. 52, no. 2, pp. 489–509, 2006.

[6] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.

CamScanner